

- **Knowledge acquisition:** Perhaps the most complicated way of deciding actions, this schema allows the robot to explore its environment and collect information relevant to its task, thereby modifying its strategy for exploration, path as well as the types of useful information stored. Algorithms to generate 'curiosity' such as Intelligent Adaptive Curiosity and Category-Based Intrinsic Motivation are used here.

Applications of robotics software - computer vision

Computer vision is an up-and-coming field wherein visual data, typically from a simple camera, is processed to generate useful information. It's been a typically 'nerdy' weekend project to hook up one's laptop camera to respond to hand gestures - wave one hand to pause a video, swipe to change tabs, and so on. With the integration of satellite data and the development of massively efficient algorithms for data processing and image processing, computer vision has seen a growth like never before.

In fact, as computer vision has so often been associated with the processing of visual data to obtain other information or features, the techniques used to process such data to determine actions of a robotic actuator have come to be known as Robot Vision. Thus, Robot Vision actually uses techniques and analyses of kinematics, references frames and locomotion in conjunction with optical sensory data. Consider the technique of Visual Servoing, which is exclusively a Robot Vision, not a Computer Vision, application. In this technique, the motion of a robot is controlled using the feedback on the robot's present position.

Robot vision has tremendous applications - it has been used in amazing laboratory demonstrations of agile quadcopters which can locate moving obstacles and move to avoid them. Similarly, it can be used by aerial robots to corroborate their location using satellite data, as well as underground or underwater robots to detect leaks, take readings from manual instruments and so on.

Applications of robotics software - control systems

Control systems is simply the field of applying optimal control signals as input to get a desired output. These outputs could be electrical, mechanical, or of any other form. Often, even the most basic of control systems involve the measurement of various input parameters, the prediction of the required next input parameters for the desired output, a measurement of the actual output, and some manner of correction terms. Often, a degree of mathematics